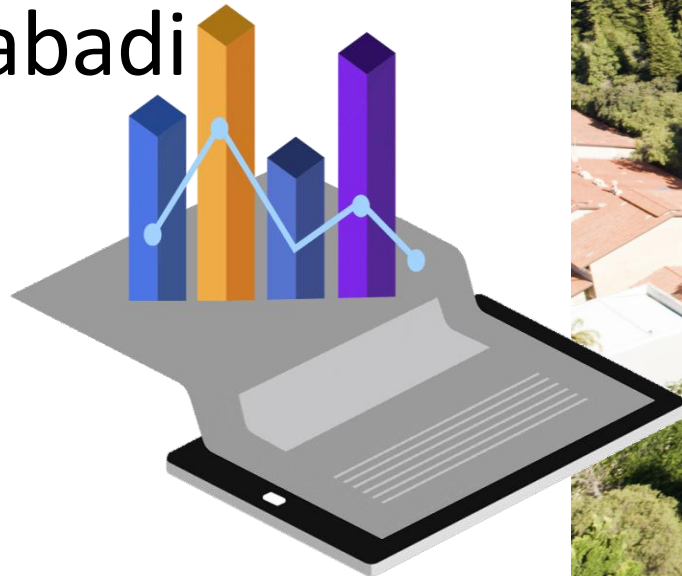


Week 5: Diagnostic Analytics in Supply Chains (SCs)

Presented by:

Dr. Mehdi Rajabi
Asadabadi



THE UNIVERSITY OF WESTERN AUSTRALIA



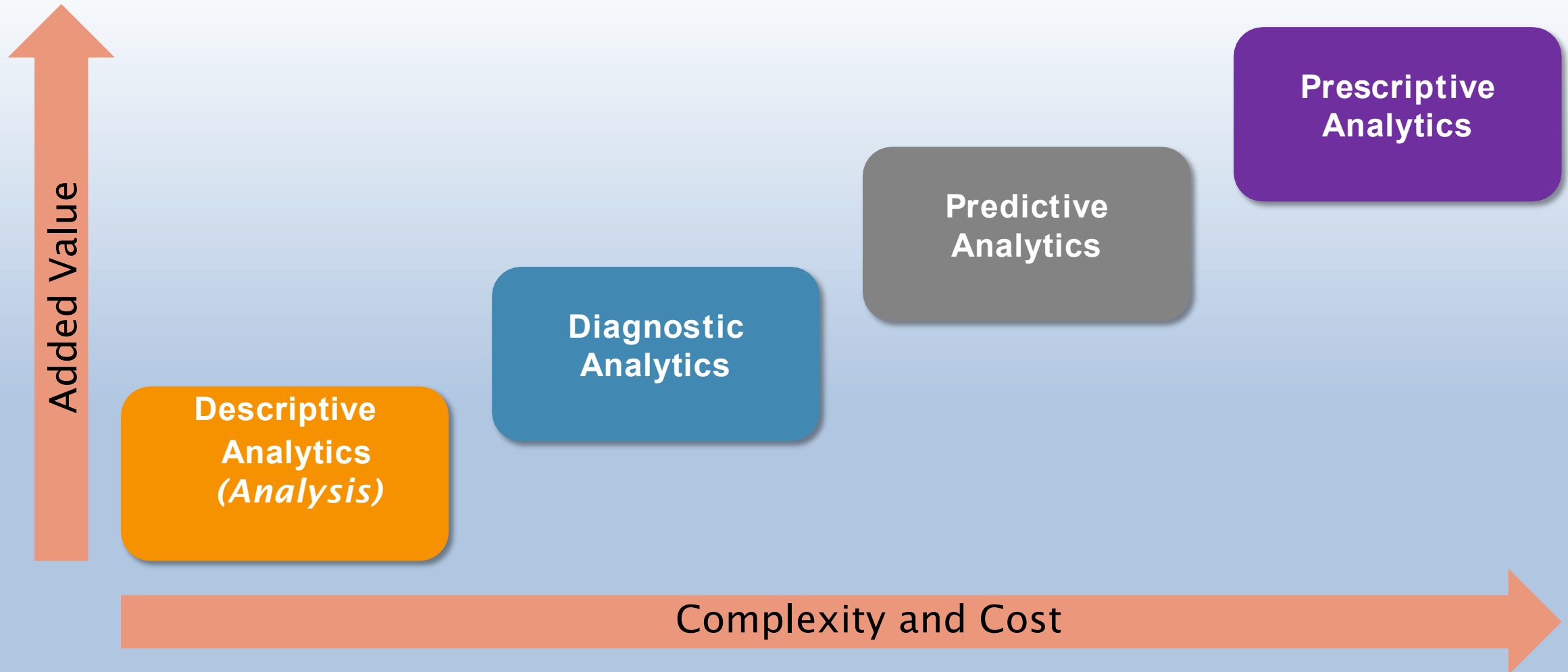
Download Analysis ToolPak

Load and activate the Analysis ToolPak (Mac)

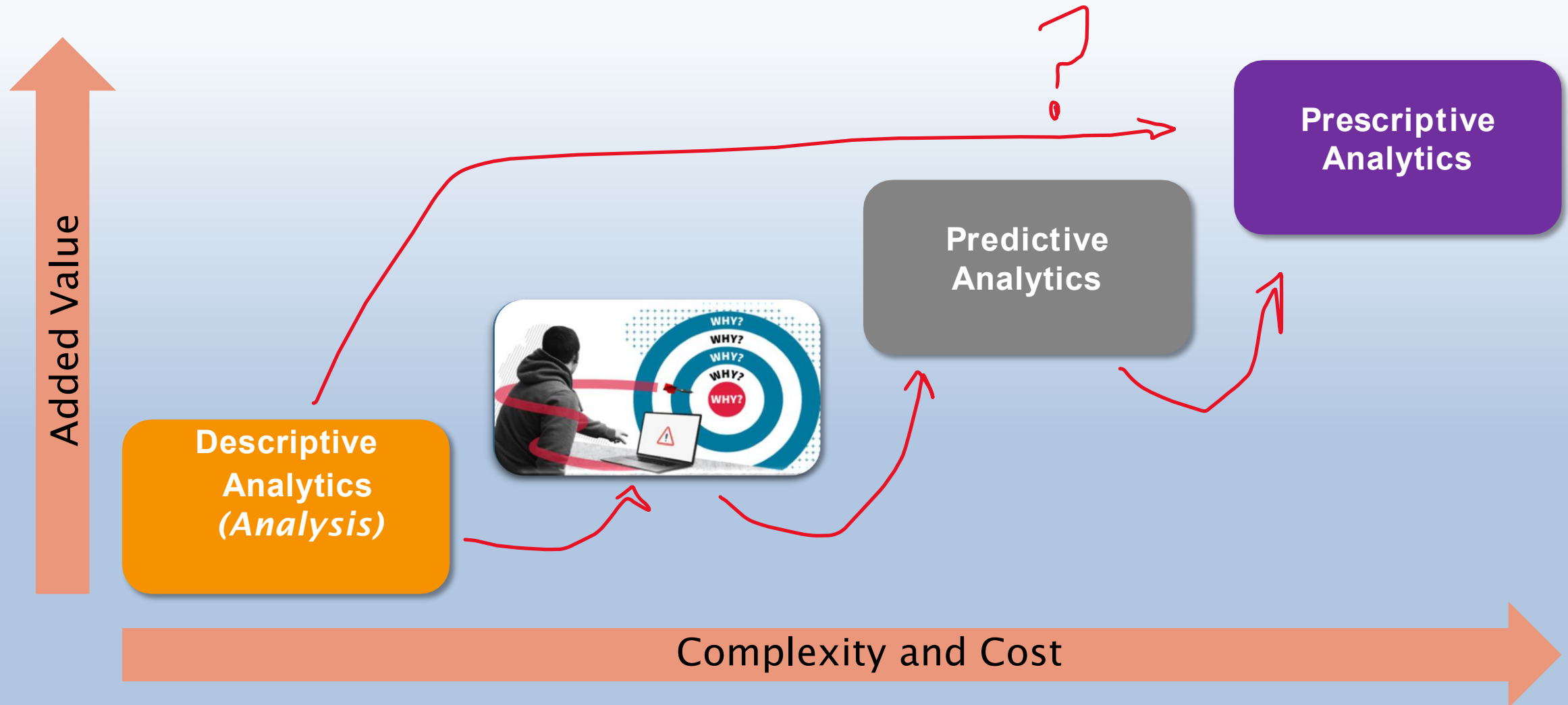
- Click the File tab, click Options, and then click the Add-Ins category. ...
- In the Manage box, select Excel Add-ins and then click Go. ...
- In the Add-Ins box, check the Analysis ToolPak check box, and then click OK.



Types of Analytics



Types of Analytics



Diagnostic Analytics

When do
we need
diagnostic
analytics?

Where someone claims something and you want to see if it is correct to make a decision.

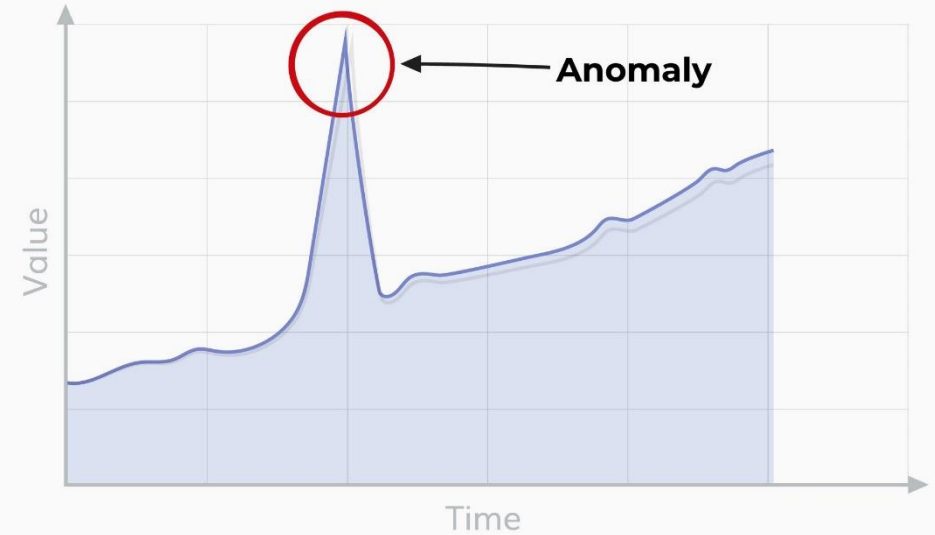
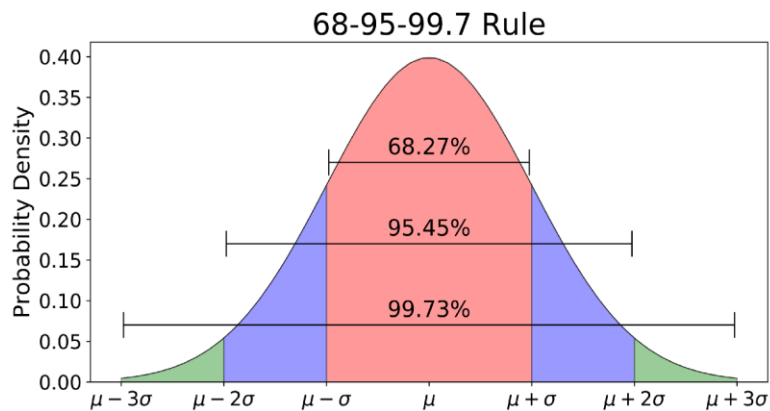
Where you notice some possible meaningful relations between data that can assist making decisions

Where there are anomalies and you do not understand why.

Diagnostic Analytics

When do we need diagnostic analytics?

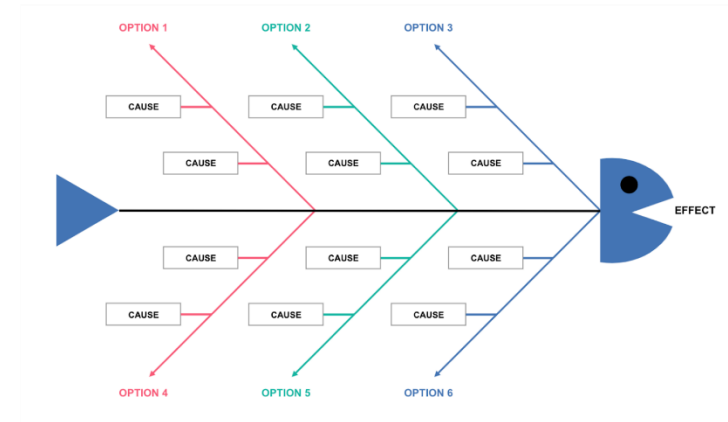
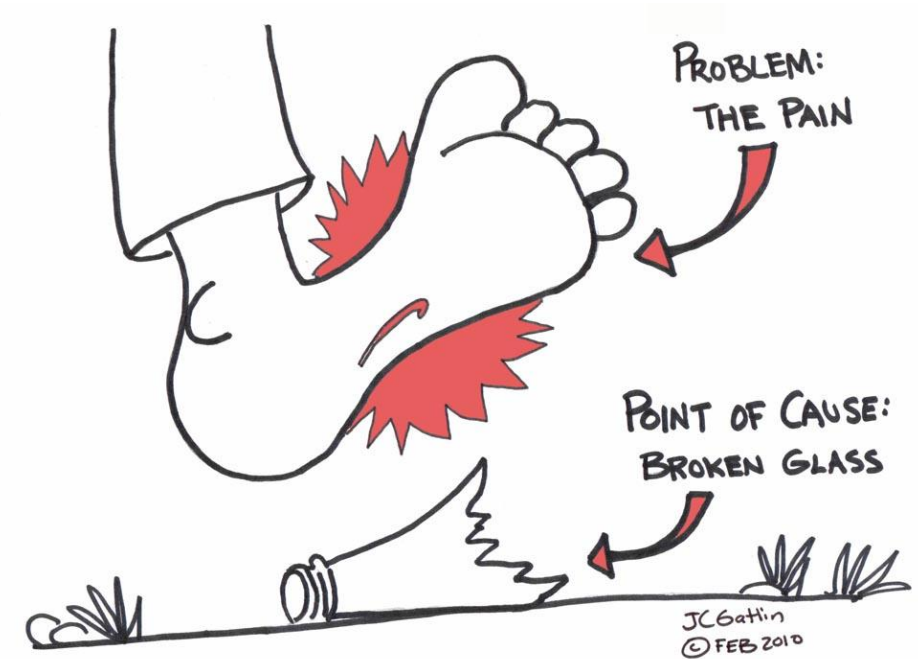
- **Anomalies**
 - ❑ Where there are **sudden changes in data, their patterns or trends**
 - ❑ Where some **specific data items are beyond what you expected**



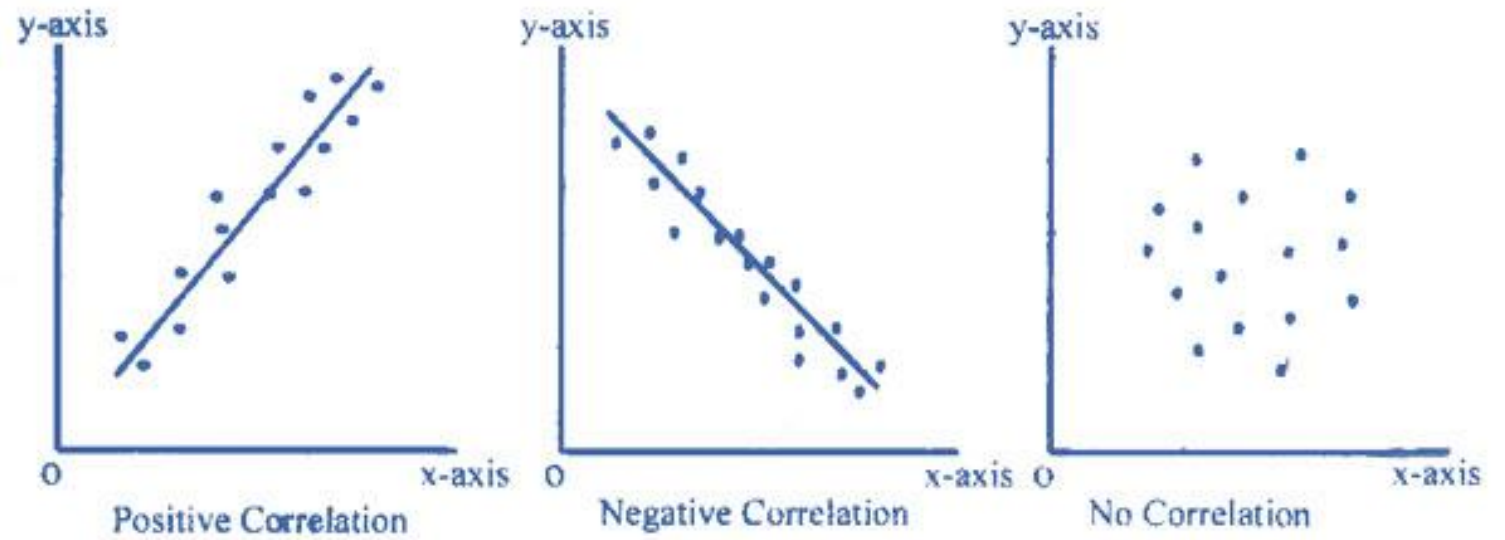
Diagnostic Analytics

Diagnostic analytics

- What to do
 - ❑ Drill into the analytics (discovery) and detect patterns
 - ❑ Determine relationships (finding the cause)



Correlation





A scenario:

- Your organisation has commenced a long-term relationship with a new supplier.
- You notice that the employee's performance is not so satisfactory in this supplier's organisation.
- In a long-term relationship, organisations often try to improve the performance of their suppliers.
- So, you set a training course for the employees of this supplier and now you want to do some further analysis of their performance.

Correlation

1. **Perfect:** If the value is near ± 1 , then it said to be a perfect correlation: as one variable increases, the other variable tends to also increase (if positive) or decrease (if negative).
2. **High degree:** If the coefficient value lies between ± 0.50 and ± 1 , then it is said to be a strong correlation.
3. **Moderate degree:** If the value lies between ± 0.30 and ± 0.49 , then it is said to be a medium correlation.
4. **Low degree:** When the value lies below $\pm .29$, then it is said to be a small correlation.

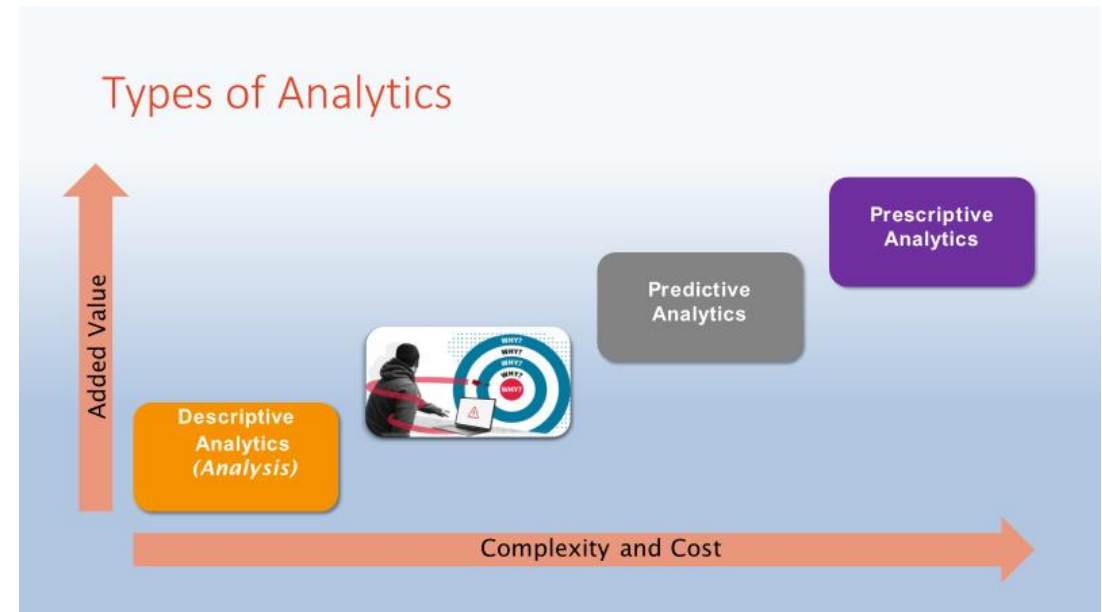
[illegible]

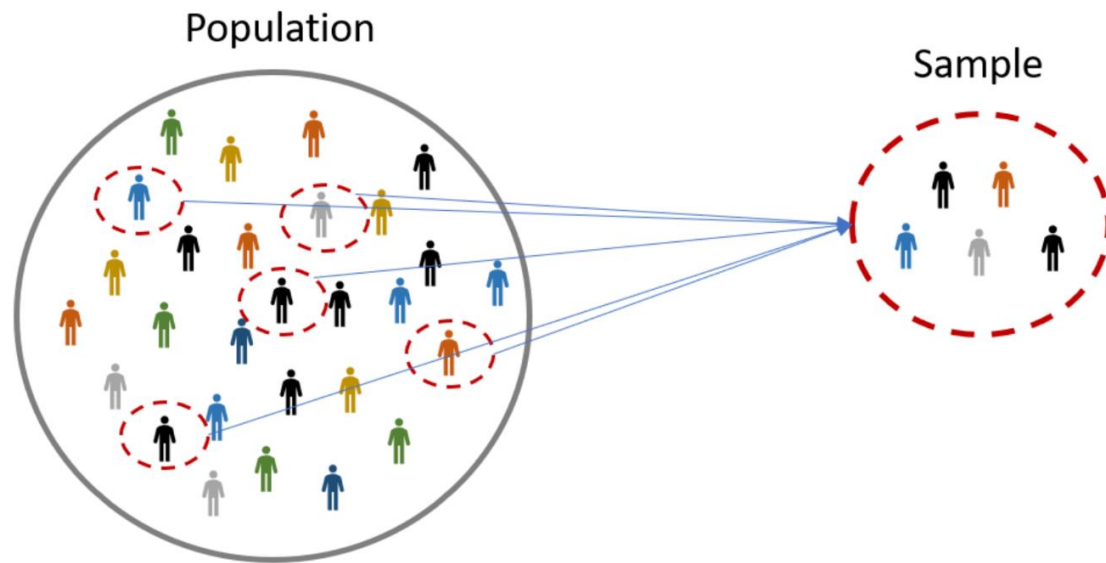
HYPOTHESIS?

Hypothesis Testing is to see whether a claim is reliable

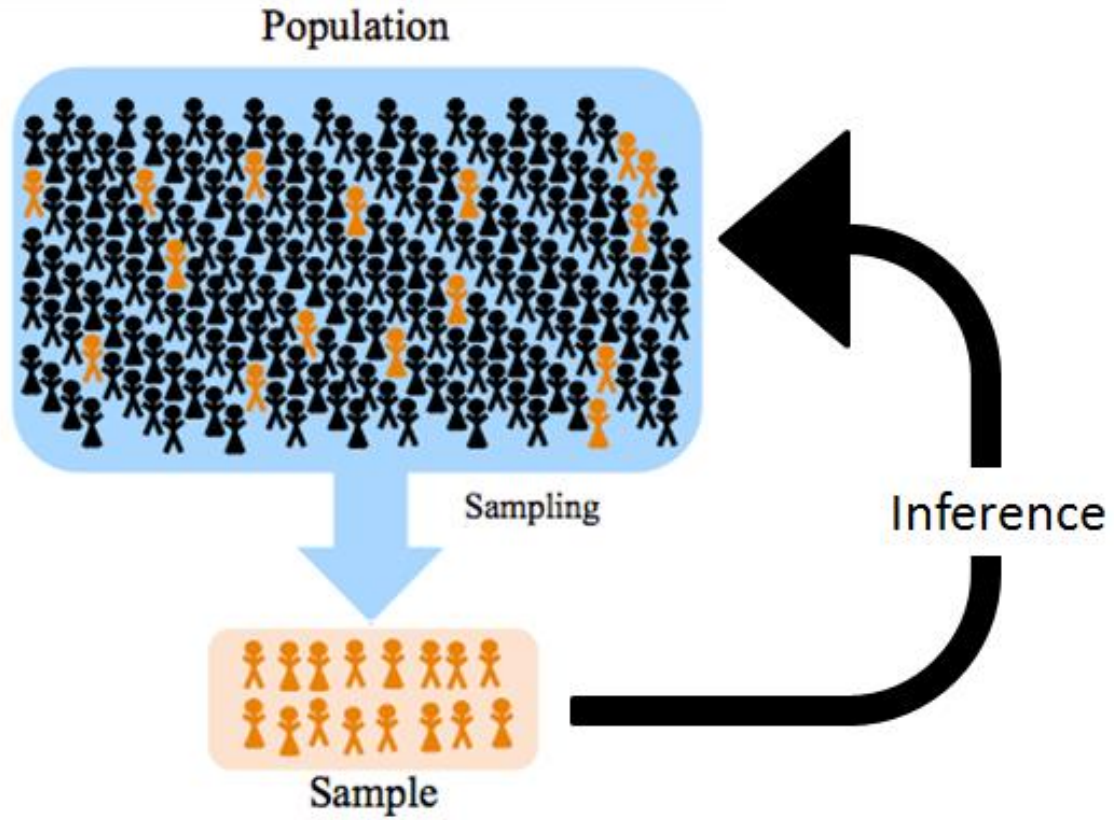
This will not only be useful in diagnostic analytics, but also if you are doing **a research study** and want to see if your claims are reliable.

This will also be quite useful for **those who are planning to transition to pursue a PhD degree**



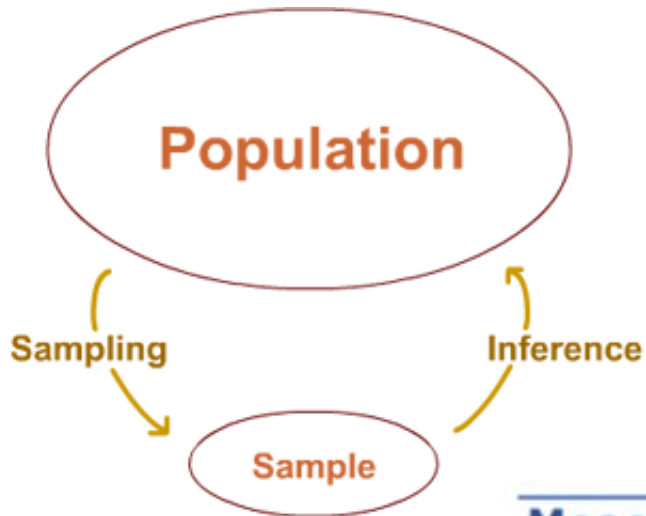


Sample vs. Population

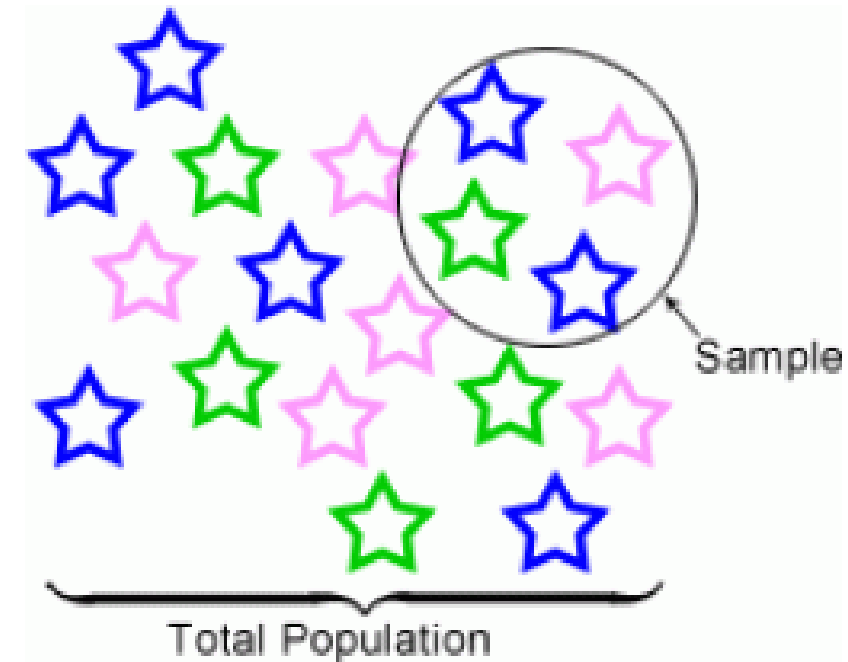


Sample vs.
Population

Statistic vs Parameter



Measurement	Sample Statistic	Population Parameter
Size	n	N
Mean	$\bar{X}$	μ
Variance	s^2	σ^2
Standard Deviation	s	σ



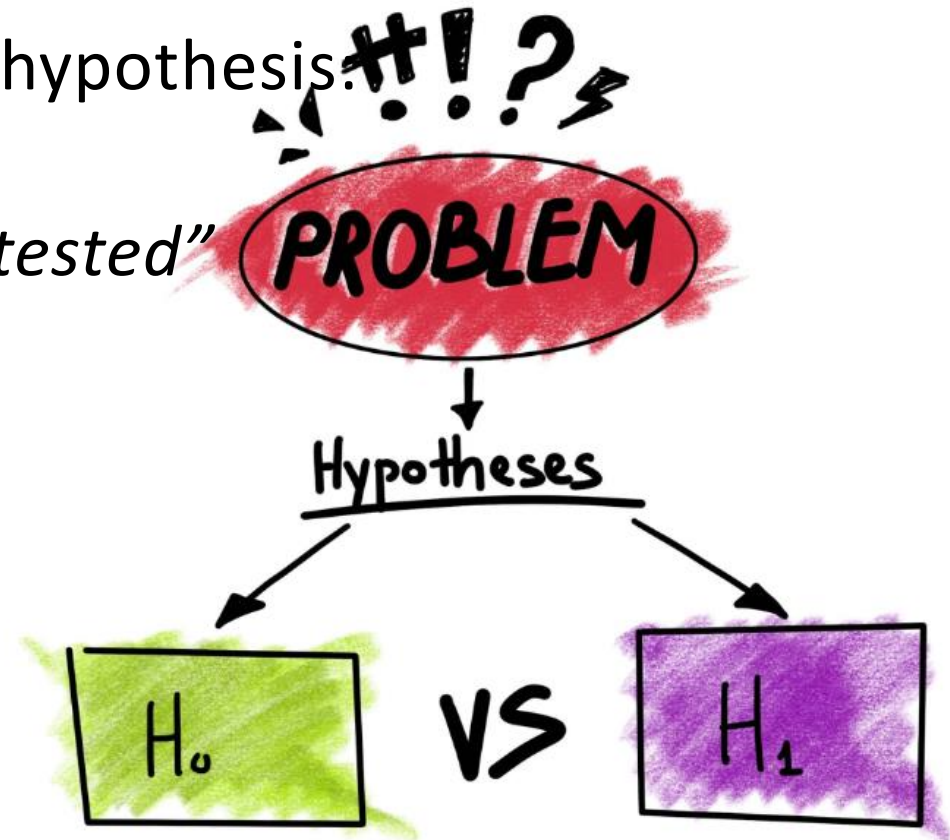
Hypothesis Testing

Informal, but easy to digest definition of a hypothesis:

“A hypothesis is an idea/claim that can be tested”

Steps to test a hypothesis:

- Formulate a hypothesis
- Find the right test
- Run the test and make a decision



Someone says the average salary of government employees in Australia is smaller than \$140,000 (claim). You have a sample saying it is \$135,000. Can you reject this claim?

Hypotheses: H0 and H1 (or Ha)

Our alternative Hypothesis (H1 or Ha):

is the claim that we like. It is the one that is pleasing, basically that is our research claim.

For H1 :

Allowed Signs: “ $\neq$ ” or “ $>$ ” or “ $<$ ”

Null Hypothesis (H0):

is the one that we want to reject (if we have enough evidence, meaning that P value is less than 0.05).

For H0 :

Allowed Signs: “ $=$ ” or “ $\geq$ ” or “ $\leq$ ”

Hypothesis Testing

Null Hypothesis

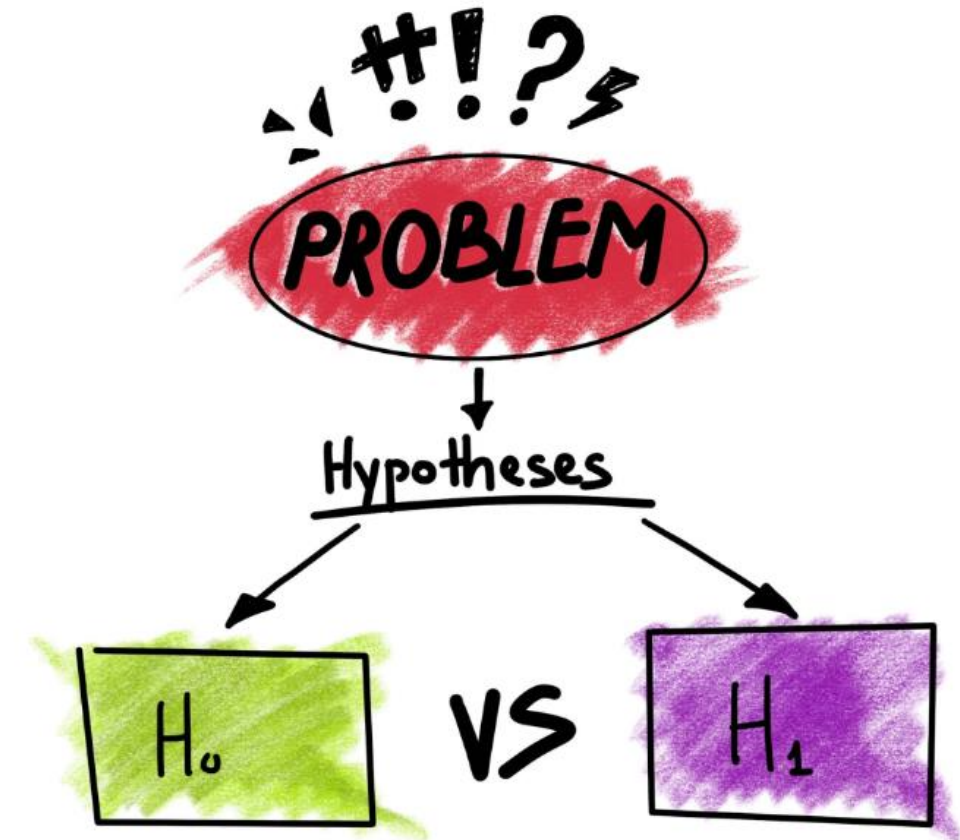
Alternative Hypothesis

H_0 : Mean Gov Employee Salaries = \$140000

H_1 : Mean Gov Employee Salaries $\neq$ \$140000

Null hypothesis is H_0

Alternative hypothesis is H_1 or H_A

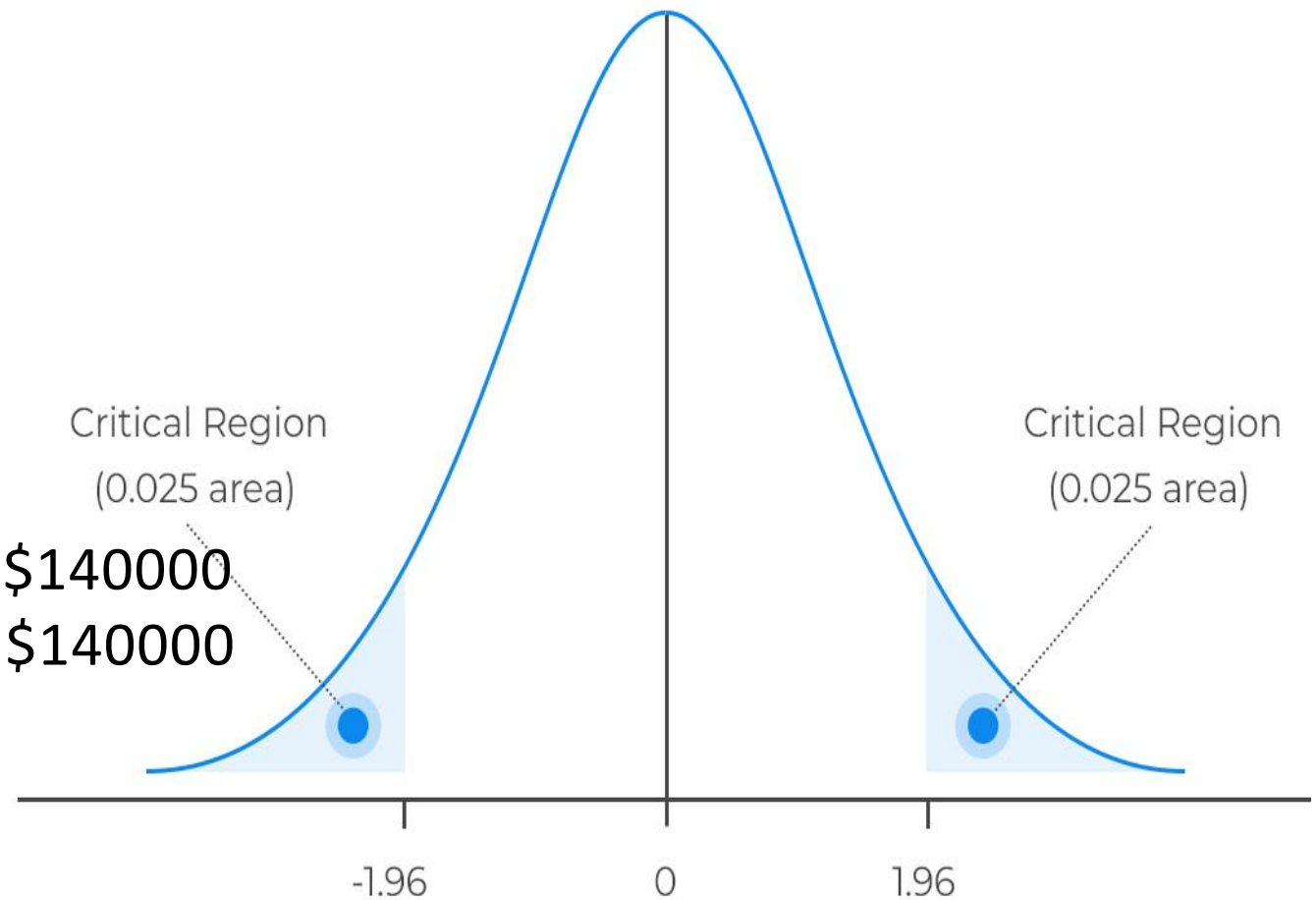


Two tailed test

Two-tailed Test

H_0 : Mean Gov Employee Salaries = \$140000

H_1 : Mean Gov Employee Salaries $\neq$ \$140000



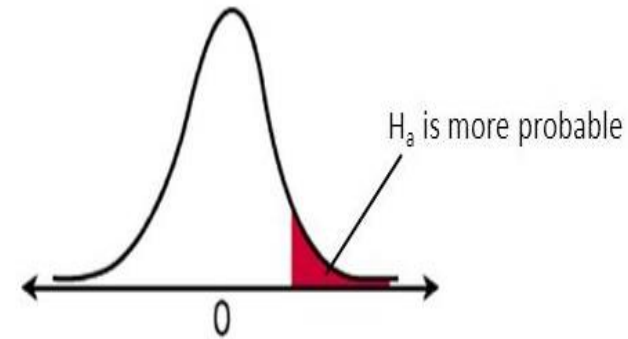
Hypothesis Testing

Null Hypothesis

Alternative Hypothesis

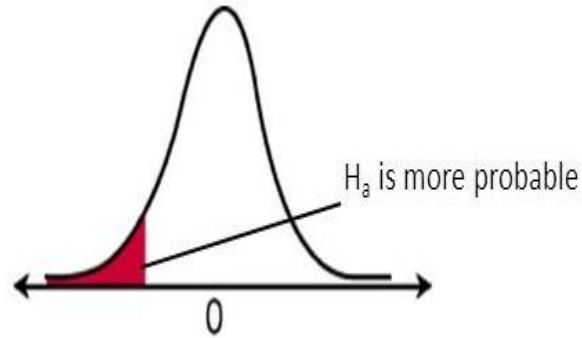
H_0 : Mean Gov Employee Salaries = \$140000

H_1 : Mean Gov Employee Salaries $\neq$ \$140000



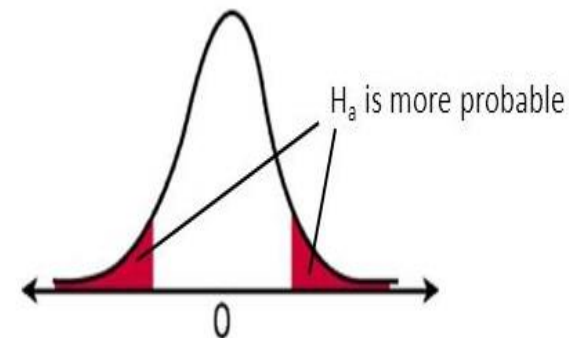
Right-tail test

$H_a: \mu > \text{value}$



Left-tail test

$H_a: \mu < \text{value}$



Two-tail test

$H_a: \mu \neq \text{value}$

Hypothesis Testing

Null Hypothesis

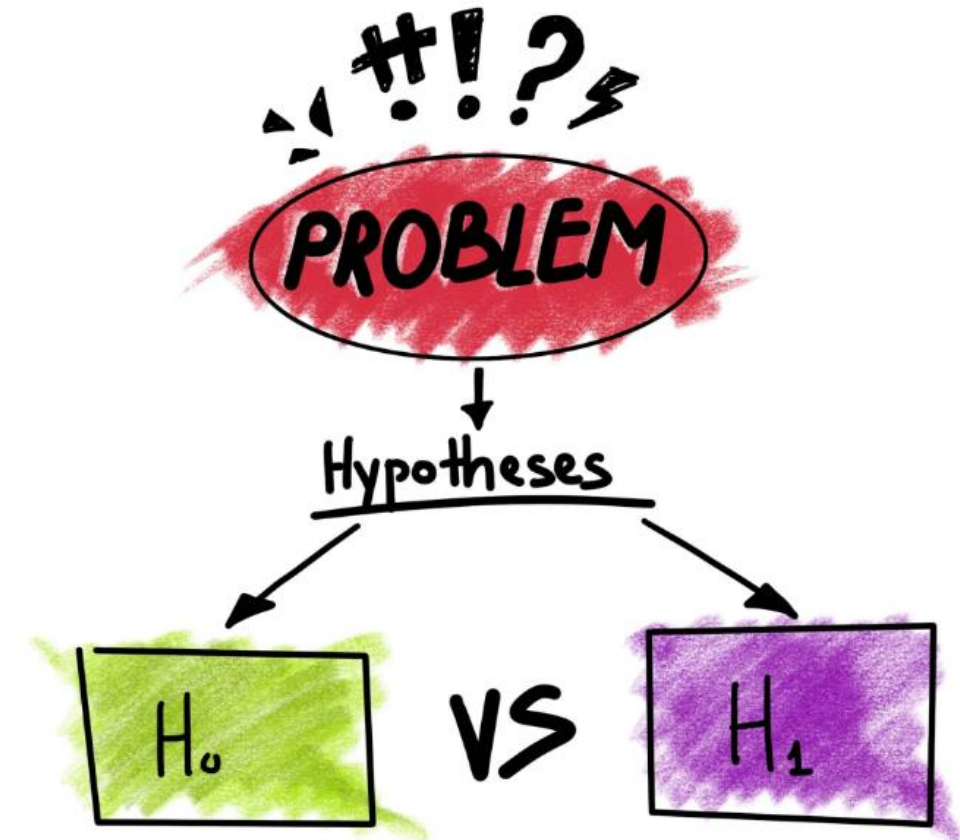
Alternative Hypothesis

H_0 : Mean Gov Employee Salaries $\leq$ 140000

H_1 : Mean Gov Employee Salaries $>$ 140000

Null hypothesis is H_0

Alternative hypothesis is H_1 or H_A

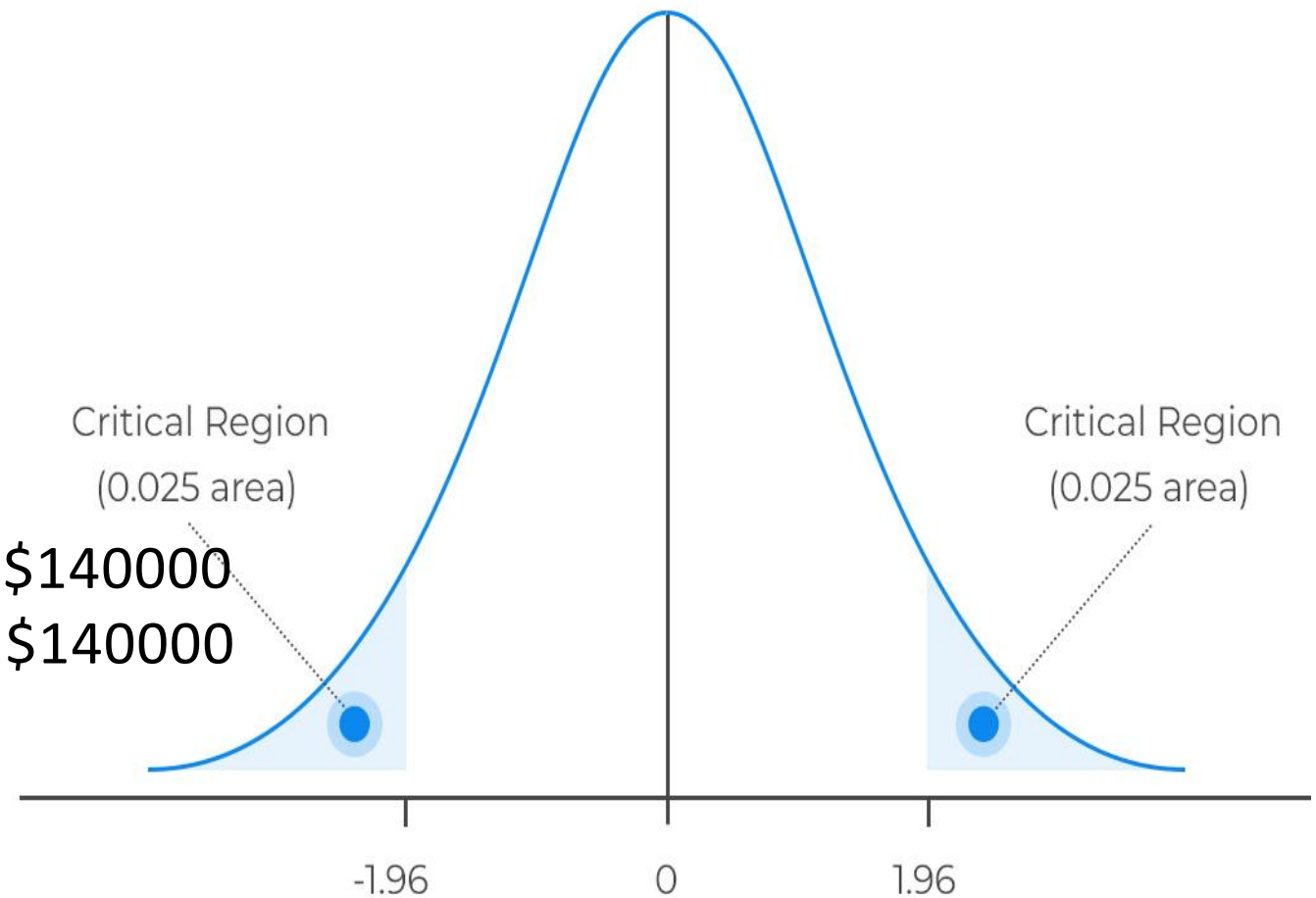


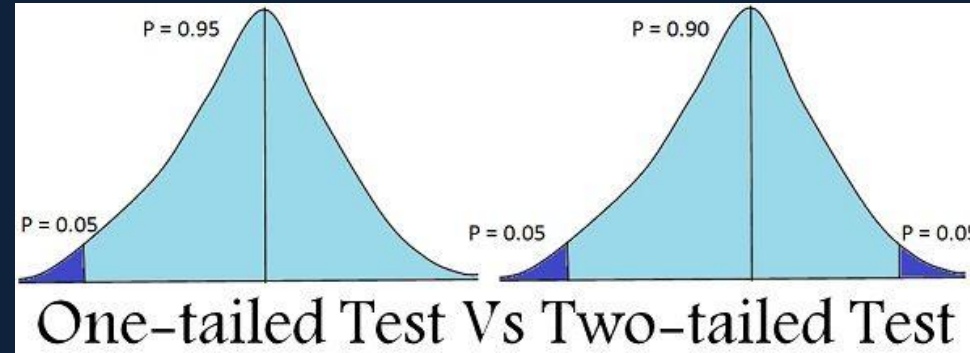
Two tailed test

Two-tailed Test

H_0 : Mean Gov Employee Salaries = \$140000

H_1 : Mean Gov Employee Salaries $\neq$ \$140000





We do not have access to data of all government job data. We **got a sample**, and the sample has the average of:

- **100,000** dollars as the average of **female salaries** and
- **98,000** as the average of **male salaries**

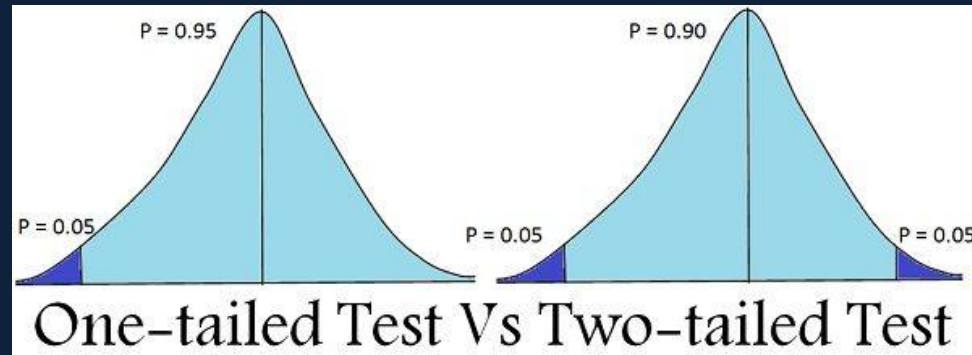
Diagnostic Analytics

We have evidence that supports our idea that female salaries are higher than males (H_1)

Can we reject H_0 that says female salaries are smaller than male salaries?

H_0 : Female salaries $\leq$ Male salary

H_1 : Female salaries $>$ Male salaries



All we can do in hypothesis testing is to reject the null hypothesis.

To reject a null hypothesis, we compute P-value, and it needs to be smaller than 5% or 0.05. Otherwise, you fail to reject H_0 .

P-value is the chance of your result being just a coincidence .

P-value of 0.05 means if we get 100 samples, the results will be supportive of rejecting the null hypothesis in 95 samples.

Steps to reject a null hypothesis

1. **Check whether data is normal** (to see which test to use) (Excel does this: Data/Data Analytics/Descriptive Statistics)
2. **Do the test** (e.g. Data/Data Analysis/t-Test), and see if you find evidence (if you find no evidence, you fail to reject the null hypothesis – no further action)
3. **If you find evidence, you need to see if whether that evidence is strong enough** to reject the null hypothesis (P-value needs to be smaller than 0.05 – P-value comes with your test results)

Normal
Distribution

Other Distributions

Examples of Parametric and non-Parametric Tests

	Parametric	Non-parametric	Example
Comparing two related samples	t-test	Wilcoxon signed rank test	Performance of a group before & after
Comparing two unrelated samples vs. a variable	t-test	Mann-Whitney U-test	Gender (M/F) vs. Job satisfaction level (Likert Scaled)
Comparing three or more related samples with one variable	ANOVA	Friedman test	Role at the company vs. Salary (low, medium, high)
Comparing three or more samples with unrelated variables	ANOVA	Kruskal-Wallis H-test	Assigned projects vs. Job satisfaction level (Likert Scale)
Comparing unrelated categories	None	Chi-square test	Company award (e.g. yes, no) vs. Performance level (high, low)
Comparing two independent ranks	Pearson product-moment correlation	Spearman rank-order test	Salary vs. Job satisfaction level (e.g. Likert Scale)

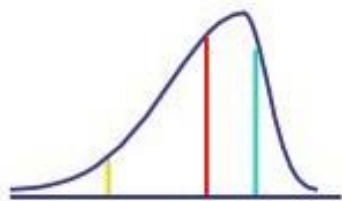
How to find out if your data is normal?

Skewness

Kurtosis

Normal Distribution Skewness

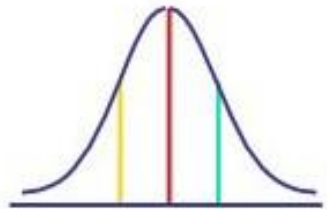
Left-Skewed



Q1 Q2 Q3



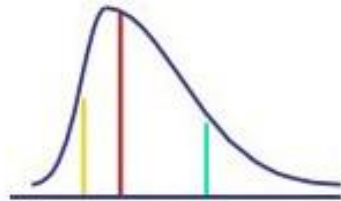
Symmetric



Q1 Q2 Q3



Right-Skewed



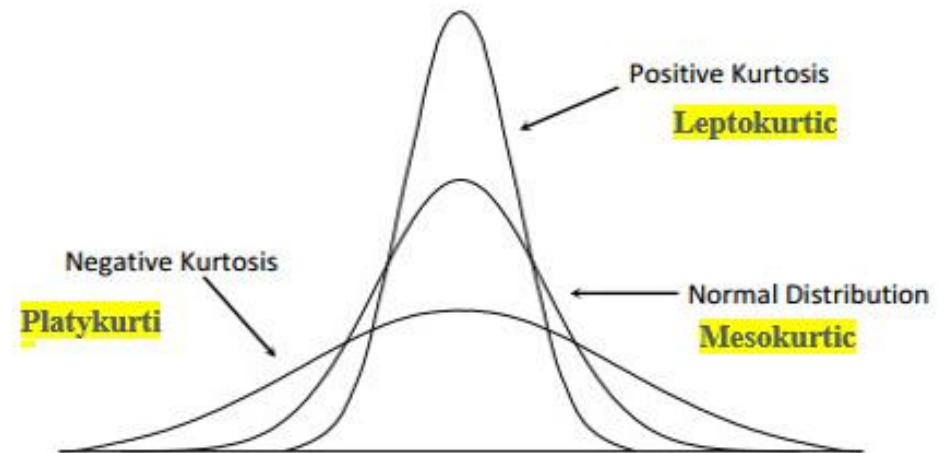
Q1 Q2 Q3



- Good if it is within -1 and 1
 - Acceptable between -2 and 2
- If two far from zero, we use nonparametric tests

Normal Distribution Kurtosis

- Good if it is within -2 to 2
 - Acceptable within -3 and 3
- If it is beyond these, we use non-parametric tests



Your Claim: Males
and female salaries
are different
(Independent
sample t-Test with
unequal variances)

Steps:

- Formulate your hypothesis
- Copy gender and salary column into a new column or sheet.
- Sort the gender column.
- Clean your data
- Separate gender data.
- Check whether you can use t-Test (or you need Wilcoxon signed rank test or Mann-Whitney U-test) (descriptive analysis – check data skewness and kurtosis)
- Click on data analysis and select 2 sample t-test assuming unequal variance
- Insert data and run the test.
- Now check the p value. If p value is less than 0.05 we reject the null hypothesis

Steps to reject a null hypothesis

1. Check whether data is normal (to see which test to use) (Excel does this: Data/Data Analytics/Descriptive Statistics)
2. Do the test (e.g. Data/Data Analysis/t-Test), and see if you find evidence (if you find no evidence, you fail to reject the null hypothesis – no further action)
3. If you find evidence, you need to see if whether that evidence is strong enough to reject the null hypothesis (P-value needs to be smaller than 0.05 – P-value comes with your test results)

Examples of Parametric and non-Parametric Tests

	Parametric	Non-parametric	Example
Comparing two related samples	t-test	Wilcoxon signed rank test	Performance of a group before & after
Comparing two unrelated samples vs. a variable	t-test	Mann-Whitney U-test	Gender (M/F) vs. Job satisfaction level (Likert Scaled)
Comparing three or more related samples with one variable	ANOVA	Friedman test	Role at the company vs. Salary (low, medium, high)
Comparing three or more samples with unrelated variables	ANOVA	Kruskal-Wallis H-test	Assigned projects vs. Job satisfaction level (Likert Scale)
Comparing unrelated categories	None	Chi-square test	Company award (e.g. yes, no) vs. Performance level (high, low)
Comparing two independent ranks	Pearson product-moment correlation	Spearman rank-order test	Salary vs. Job satisfaction level (e.g. Likert Scale)

Your Second
Claim: Male
salaries are
smaller than
female salaries

Steps:

- Formulate your hypothesis
- Copy gender and salary column into a new column or sheet.
- Sort the gender column.
- Separate gender data.
- Check whether you can use t-Test (or you need Wilcoxon signed rank test or Mann-Whitney U-test) (descriptive analysis – check data skewness and kurtosis)
- Click on data analysis and select 2 sample t-test assuming unequal variance
- Insert data and run the test.
- Now check the p value. If p value is less than 0.05 we reject the null hypothesis

E	F	G
t-Test: Two-Sample Assuming Unequal Variances		
	<i>Variable 1</i>	<i>Variable 2</i>
Mean	33	24.8
Variance	160	21.7
Observations	6	5
Hypothesized Mean Difference	0	
df	7	
t Stat	1.47260514	
P(T<=t) one-tail	0.092170202	
t Critical one-tail	1.894578605	
P(T<=t) two-tail	0.184340405	
t Critical two-tail	2.364624252	

t-Test: Two-Sample Assuming Unequal Variances		
	<i>Salary</i>	<i>Salary</i>
Mean	104420	110347.8261
Variance	530534285.7	604339226
Observations	50	92
Hypothesized Mean	0	
df	106	
t Stat	-1.430174571	
P(T<=t) one-tail	0.077803996	
t Critical one-tail	1.659356034	
P(T<=t) two-tail	0.155607992	
t Critical two-tail	1.982597262	

Claim: Female salaries are smaller than Male Salaries

- Can you formulate this on your Excel sheet:

H0: Average of Female Salaries $\geq$ Average of male Salaries

H1: Average of Female Salaries $<$ Average of male Salaries

Your Claim:
Female salaries
are smaller than
Male Salaries

H0: Average of Female Salaries $\geq$ Average of male Salaries

H1: Average of Female Salaries $<$ Average of male Salaries

t-Test: Two-Sample Assuming Unequal Variances

	<i>Female Salary</i>	<i>Male Salary</i>
Mean	104420	110347.8261
Variance	530534285.7	604339226
Observations	50	92
Hypothesized Mean	0	
df	106	
t Stat	-1.430174571	
P(T<=t) one-tail	0.077803996	
t Critical one-tail	1.659356034	
P(T<=t) two-tail	0.155607992	
t Critical two-tail	1.982597262	

H0: Average of Female Salaries $\geq$ Average of male Salaries
H1: Average of Female Salaries $<$ Average of male Salaries

t-Test: Two-Sample Assuming Unequal Variances

	<i>Female Salary</i>	<i>Male Salary</i>
Mean	104420	110347.8261
Variance	530534285.7	604339226
Observations	50	92
Hypothesized Mean	0	
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t Critical one-tail	1.659356034	
P(T<=t) two-tail	0.155607992	
t Critical two-tail	1.982597262	

Do we have evidence?

Yes

Is it strong enough?

No (one-tail P-value is not smaller than 0.05)

H0: Average of Female Salaries $\geq$ Average of male Salaries
H1: Average of Female Salaries $<$ Average of male Salaries

Claim: The average working hours of employees is more than 40 hours

Working Hours

Descriptive Statistics

Input

Input Range: SCL:SCL

Grouped By: ☒ Columns ☐ Rows

☒ Labels in First Row

Output options

☐ Output Range:

☒ New Worksheet Ply:

☐ New Workbook

☒ Summary statistics

☐ Confidence Level for Mean: 95 %

☐ Kth Largest: 1

☐ Kth Smallest: 1

OK Cancel Help

Working Hours	
Mean	44.88028169
Standard Error	0.710768969
Median	42
Mode	35
Standard Deviation	8.469789781
Sample Variance	71.73733893
Kurtosis	-1.179444628
Skewness	0.466679585
Range	25
Minimum	35
Maximum	60
Sum	6373
Count	142

ANOVA

t-Test was for two sets/groups/columns of data. An **ANOVA** test is a way to find out difference between more than two testing groups. In other words, they help you to figure out if you need to reject the null hypothesis.

Examples of testing more than two groups:

A group of psychiatric patients are trying three different therapies: counselling, medication and biofeedback. You want to see if one therapy is better than the others.

A manufacturer has three different processes to make light bulbs. They want to know if one process is better than the other.

Students from different colleges take the same exam. You want to see if one college outperforms the other.

E	F	G	H	I	J	K
Anova: Single Factor						
SUMMARY						
<i>Groups</i>	<i>Count</i>	<i>Sum</i>	<i>Average</i>	<i>Variance</i>		
Column 1	9	435	48.33333	23.5		
Column 2	7	420	60	32.33333		
Column 3	9	393	43.66667	50.5		
ANOVA						
<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	1085.84	2	542.92	15.19623	7.16E-05	3.443357
Within Groups	786	22	35.72727			
Total	1871.84	24				



CEO changed and assigned new managers
to different production lines

Claim: The employee satisfactory levels of
some lines have dropped.

Formulating the hypothesis

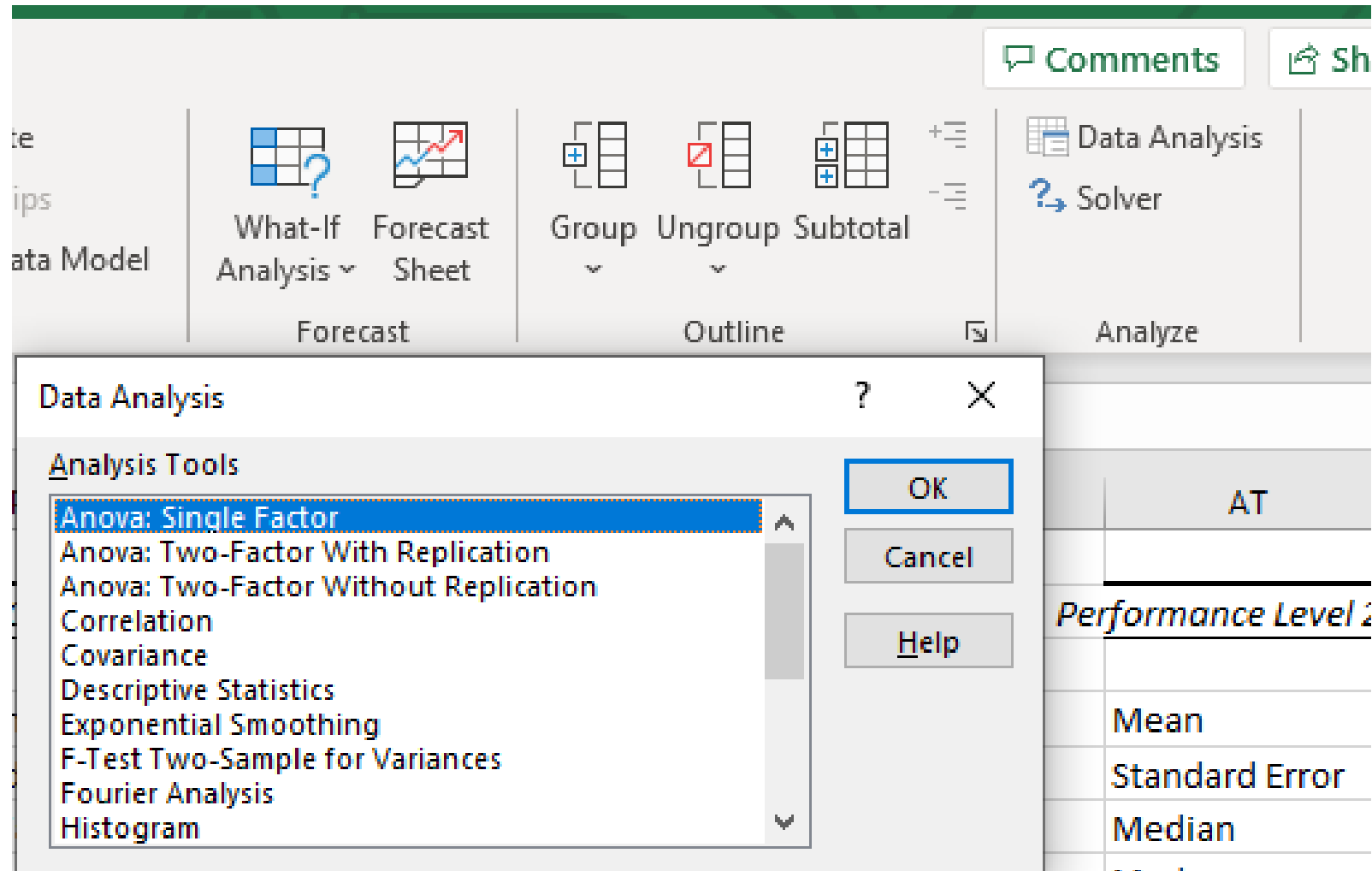
H0:

These groups are similar

H1:

These groups are not similar

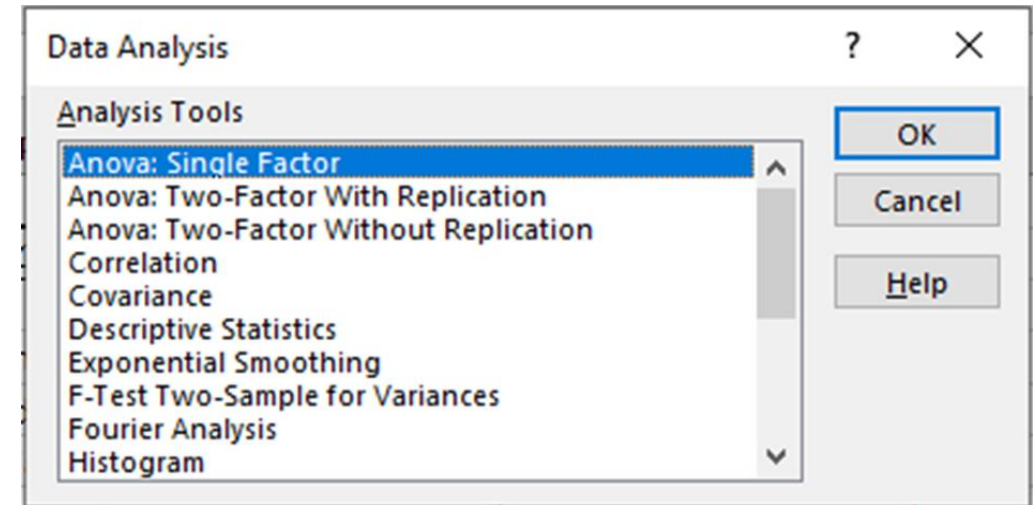
Data/Data Analysis/ANOVA/OK



H0: These groups are similar

H1: These groups are not similar

Why did we select
single factor!?



1. Why do certain products have consistently low ending inventory?
You can check 'Ending Inventory' and check if high sales volume is a factor.
2. Which products frequently experience stock replenishment?
Analyse the frequency of non-zero 'Received Inventory' entries for each product.
3. Are there any patterns of inventory shortage leading to stockouts?
Correlate low 'Ending Inventory' levels with instances of stockouts (derived from Sales Data).
4. What causes fluctuations in inventory levels for specific stores?
Compare inventory levels with sales data and external factors like promotions or seasonal trends.
5. Investigate the -relationship between inventory levels and discounts offered.
Analyze if higher 'Discount Applied' (from Sales Data) correlates with lower 'Ending Inventory'.

Examples of Diagnostic Analytics in Inventory Data

1. Why do some suppliers have lower quality ratings?
Investigate factors contributing to lower quality scores among suppliers.
2. Are there relations between cost and quality rating.
Determine if higher costs correlate with better quality.
3. Why do certain stores have higher waste levels?
Correlate waste levels with other factors like delivery delays or transportation costs.
4. Determine the cause of any significant spikes in transportation costs.
Examine the months with high transportation costs and cross-reference with external events.
5. What factors contribute to the variance in risk factor scores?
Explore the variation in risk factors to understand underlying causes.
6. Why are sale levels fluctuating; can it be related to discounts?
Assess the impact of discounts on sales volumes.
Compare 'Units Sold' with 'Discount Applied' to see correlation between discounts and sales.
7. Analyze if higher-priced products tend to sell less.
Correlate 'Sale Price' with 'Units Sold' across products.
8. Claim: sales volumes are affected by store location.
Compare 'Units Sold' across 'Store Name' and analyze differences.

Examples of Diagnostic Analytics in Suppliers And Sales Data

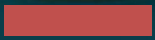
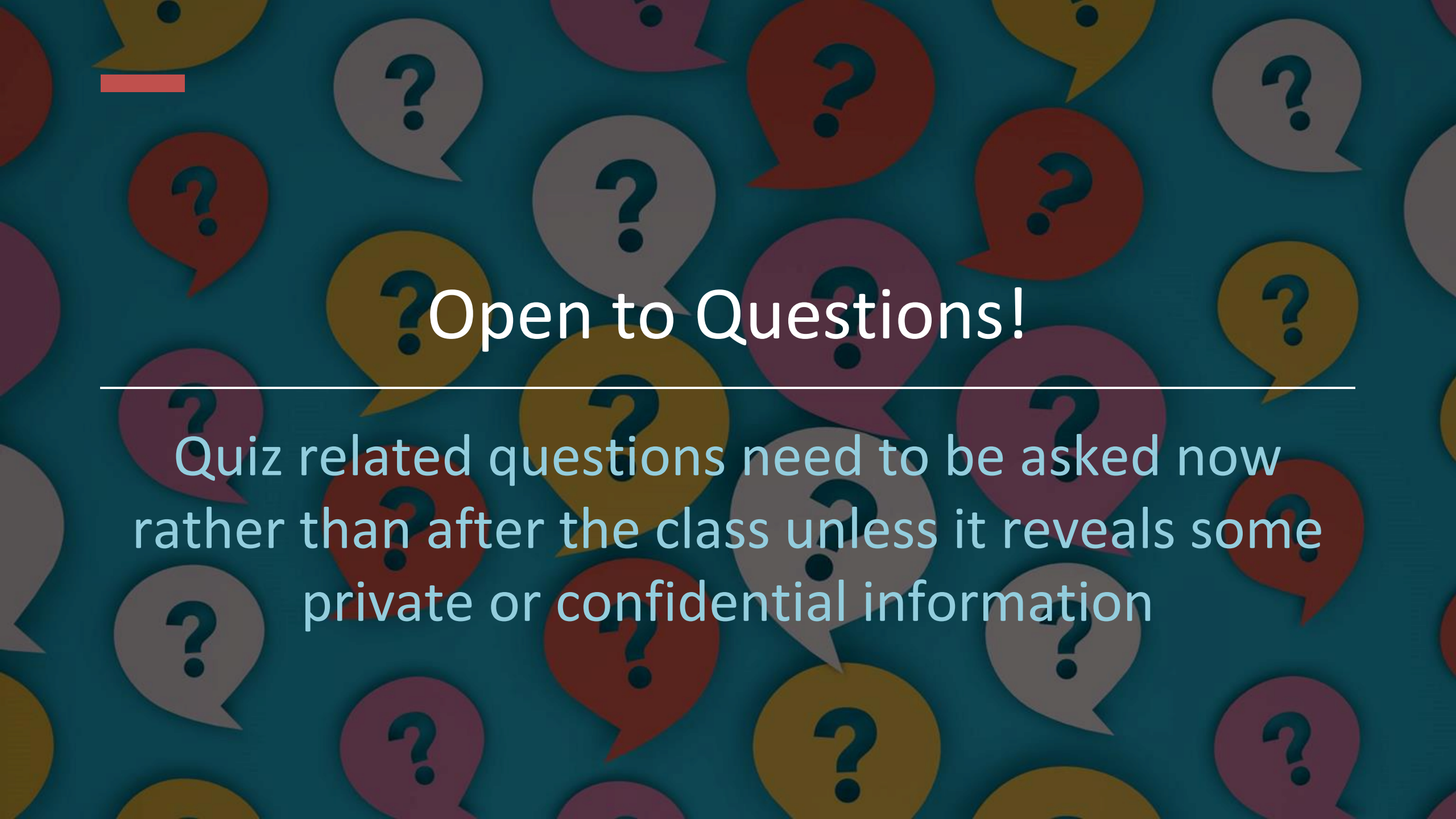
Important Notes



The quiz will be paper based.

You need to be in the class.

Note: Someone will be checking your IDs.



Open to Questions!

Quiz related questions need to be asked now rather than after the class unless it reveals some private or confidential information